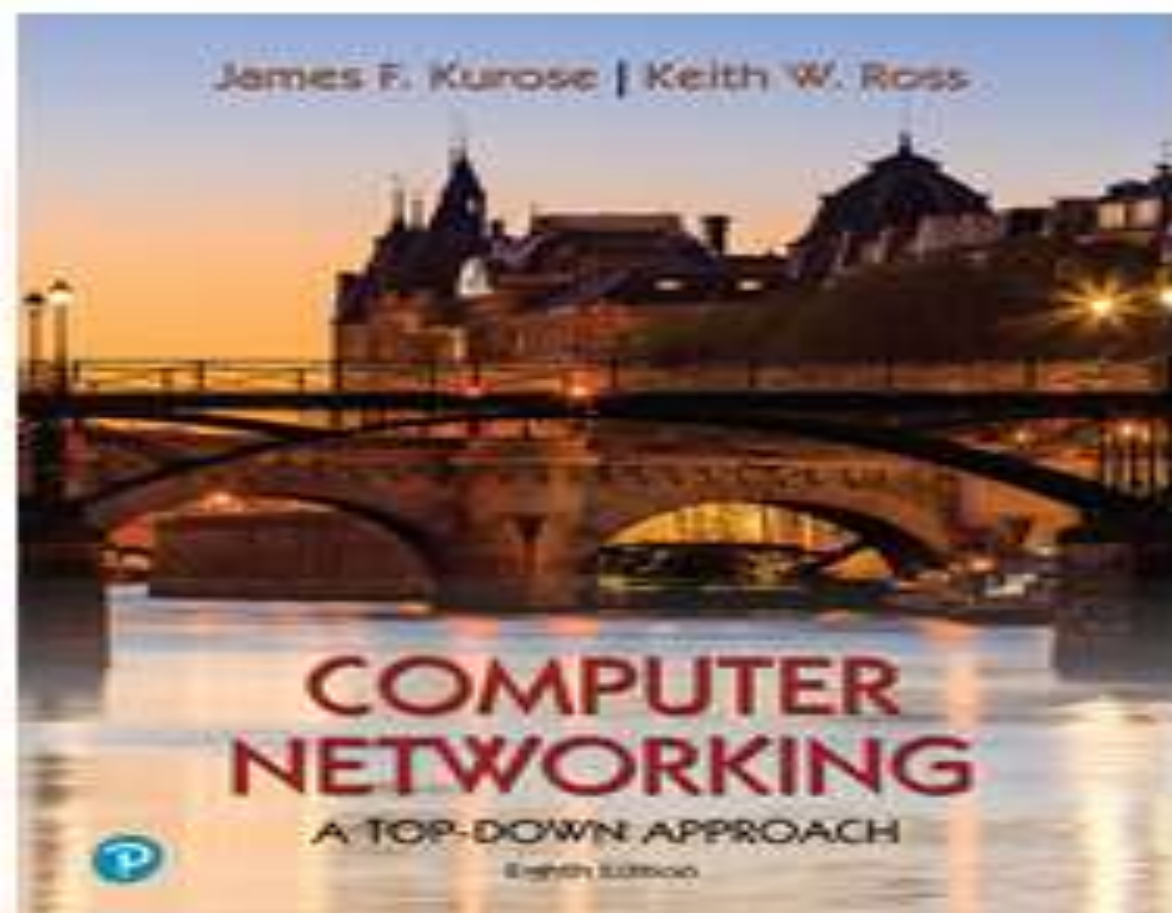




Computer Networking: A Top-Down Approach

8th edition

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Chapter 1 Computer Networks and the Internet

1.1 What Is the Internet?

In this book, we'll use the public Internet, a specific computer network, as our principal vehicle for discussing computer networks and their protocols.

But what *is* the Internet? There are a couple of ways to answer this question.

First, we can describe the nuts and bolts of the Internet, that is, the basic hardware and software components that make up the Internet.

Second, we can describe the Internet in terms of a networking infrastructure that provides services to distributed applications.

1.1.1 A Nuts-and-Bolts Description

The Internet is a computer network that interconnects billions of computing devices throughout the world.

Not too long ago, these computing devices were primarily traditional desktop computers, Linux workstations, and so-called servers that store and transmit information such as Web pages and e-mail messages.

Increasingly, however, users connect to the Internet with smartphones and tablets—today, close to half of the world’s population are active mobile Internet users.

Furthermore, non-traditional Internet “things” such as TVs, gaming consoles, thermostats, home security systems, home appliances, watches, eye glasses, cars, traffic control systems, and more are being connected to the Internet.

In Internet jargon, all of these devices are called **hosts** or **end systems**.

By some estimates, there were about 18 billion devices connected to the Internet in 2017, and the number will reach 28.5 billion by 2022 [Cisco VNI 2020].

End systems are connected together by a network of **communication links** and **packet switches**.

Different links can transmit data at different rates, with the **transmission rate** of a link measured in bits/second.

When one end system has data to send to another end system, the sending end system segments the data and adds header bytes to each segment.

The resulting packages of information, known as **packets** in the jargon of computer networks, are then sent through the network to the destination end system, where they are reassembled into the original data.

A packet switch takes a packet arriving on one of its incoming communication links and forwards that packet on one of its outgoing communication links.

Packet switches come in many shapes and flavors, but the two most prominent types in today's Internet are **routers** and **link-layer switches**.

Both types of switches forward packets toward their ultimate destinations.

Link-layer switches are typically used in access networks, while routers are typically used in the network core.

The sequence of communication links and packet switches traversed by a packet from the sending end system to the receiving end system is known as a **route** or **path** through the network.

End systems access the Internet through **Internet Service Providers (ISPs)**, including

- residential ISPs such as local cable or telephone companies;
- corporate ISPs;
- university ISPs;
- ISPs that provide WiFi access in airports, hotels, coffee shops, and other public places; and
- cellular data ISPs, providing mobile access to our smartphones and other devices.

Each ISP is in itself a network of packet switches and communication links. ISPs also provide Internet access to content providers, connecting servers directly to the Internet.

The Internet is all about connecting end systems to each other, so the ISPs that provide access to end systems must also be interconnected.

These lower-tier ISPs are thus interconnected through national and international upper-tier ISPs and these upper-tier ISPs are connected directly to each other.

An upper-tier ISP consists of high-speed routers interconnected with high-speed fiber-optic links.

Each ISP network, whether upper-tier or lower-tier, is managed independently,

End systems, packet switches, and other pieces of the Internet run **protocols** that control the sending and receiving of information within the Internet.

The **Transmission Control Protocol (TCP)** and the **Internet Protocol (IP)** are two of the most important protocols in the Internet.

The Internet's principal protocols are collectively known as **TCP/IP**.

Internet standards are developed by the Internet Engineering Task Force (IETF) [IETF 2020].

The IETF standards documents are called **requests for comments (RFCs)**. RFCs started out as general requests for comments to resolve network and protocol design problems that faced the precursor to the Internet [Allman 2011].

RFCs tend to be quite technical and detailed.

They define protocols such as TCP, IP, HTTP (for the Web), and SMTP (for e-mail).

Other bodies also specify standards for network components, most notably for network links.

The IEEE 802 LAN Standards Committee [IEEE 802 2020], for example, specifies the Ethernet and wireless WiFi standards.

1.1.2 A Services Description

But we can also describe the Internet from an entirely different angle—namely, as *an infrastructure that provides services to applications*.

In addition to traditional applications such as e-mail and Web surfing, Internet applications include mobile smartphone and tablet applications, including Internet messaging, mapping with real-time road-traffic information, music streaming movie and television streaming, online social media, video conferencing, multi-person games, and location-based recommendation systems.

The applications are said to be **distributed applications**, since they involve multiple end systems that exchange data with each other.

Importantly, Internet applications run on end systems—they do not run in the packet switches in the network core.

Because applications run on end systems, you are going to need to write programs that run on the end systems.

You might, for example, write your programs in Java, C, or Python.

Now, because you are developing a distributed Internet application, the programs running on the different end systems will need to send data to each other.

And here we get to a central issue—one that leads to the alternative way of describing the Internet as a platform for applications.

How does one program running on one end system instruct the Internet to deliver data to another program running on another end system?

End systems attached to the Internet provide a **socket interface** that specifies how a program running on one end system asks the Internet infrastructure to deliver data to a specific destination program running on another end system. This Internet socket interface is a set of rules that the sending program must follow so that the Internet can deliver the data to the destination program.

the Internet provides multiple services to its applications. When you develop an Internet application, you too must choose one of the Internet's services for your application.

1.1.3 What Is a Protocol?

A Human Analogy

Consider what you do when you want to ask someone for the time of day.

Human protocol (or good manners, at least) dictates that one first offer a greeting (the first “Hi”) to initiate communication with someone else.

The typical response to a “Hi” is a returned “Hi” message.

Implicitly, one then takes a cordial “Hi” response as an indication that one can proceed and ask for the time of day.

A different response to the initial “Hi” (such as “Don’t bother me!” or “I don’t speak English,” or some unprintable reply) might indicate an unwillingness or inability to communicate.

In this case, the human protocol would be not to ask for the time of day.

Sometimes one gets no response at all to a question, in which case one typically gives up asking that person for the time.

Note that in our human protocol, *there are specific messages we send, and specific actions we take in response to the received reply messages or other events* (such as no reply within some given amount of time).

Clearly, transmitted and received messages, and actions taken when these messages are sent or received or other events occur, play a central role in a human protocol.

Network Protocols

A network protocol is similar to a human protocol, except that the entities exchanging messages and taking actions are hardware or software components of some device (for example, computer, smartphone, tablet, router, or other network-capable device).

All activity in the Internet that involves two or more communicating remote entities is governed by a protocol.

Protocols are running everywhere in the Internet, and consequently much of this book is about computer network protocols.

As an example of a computer network protocol with which you are probably familiar, consider what happens when you make a request to a Web server, that is, when you type the URL of a Web page into your Web browser.

First, your computer will send a connection request message to the Web server and wait for a reply. The Web server will eventually receive your connection request message and return a connection reply message.

Knowing that it is now OK to request the Web document, your computer then sends the name of the Web page it wants to fetch from that Web server in a GET message.

Finally, the Web server returns the Web page (file) to your computer.

*A **protocol** defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event.*

1.2 The Network Edge

We begin in this section at the edge of the network and look at the components with which we are most familiar—namely, the computers, smartphones and other devices that we use on a daily basis.

Recall from the previous section that in computer networking jargon, the computers and other devices connected to the Internet are often referred to as end systems.

They are referred to as end systems because they sit at the edge of the Internet. The Internet's end systems include desktop computers (e.g., desktop PCs, Macs, and Linux boxes), servers (e.g., Web and e-mail servers), and mobile devices (e.g., laptops, smartphones, and tablets).

End systems are also referred to as *hosts* because they host (that is, run) application programs such as a Web browser program, a Web server program, an e-mail client program, or an e-mail server program.

Throughout this book we will use the terms hosts and end systems interchangeably; that is, *host = end system*.

Hosts are sometimes further divided into two categories: **clients** and **servers**. Informally, clients tend to be desktops, laptops, smartphones, and so on, whereas servers tend to be more powerful machines that store and distribute Web pages, stream video, relay e-mail, and so on.

Today, most of the servers from which we receive search results, e-mail, Web pages, videos and mobile app content reside in large **data centers**.

1.2.1 Access Networks

let's next consider the access network—the network that physically connects an end system to the first router (also known as the “edge router”) on a path from the end system to any other distant end system.

Home Access: DSL, Cable, FTTH, and 5G Fixed Wireless

let's begin our overview of access networks by considering how homes connect to the Internet.

Today, the two most prevalent types of broadband residential access are **digital subscriber line (DSL)** and cable.

A residence typically obtains DSL Internet access from the same local telephone company (telco) that provides its wired local phone access.

Thus, when DSL is used, a customer's telco is also its ISP.

each customer's DSL modem uses the existing telephone line exchange data with a digital subscriber line access multiplexer (DSLAM) located in the telco's local central office (CO).

The home's DSL modem takes digital data and translates it to high- frequency tones for transmission over telephone wires to the CO; the analog signals from many such houses are translated back into digital format at the DSLAM.

The residential telephone line carries both data and traditional telephone signals simultaneously, which are encoded at different frequencies:

On the customer side, a splitter separates the data and telephone signals arriving to the home and forwards the data signal to the DSL modem.

On the telco side, in the CO, the DSLAM separates the data and phone signals and sends the data into the Internet.

Hundreds or even thousands of households connect to a single DSLAM.

Engineers have expressly designed DSL for short distances between the home and the CO; generally, if the residence is not located within 5 to 10 miles of the CO, the residence must resort to an alternative form of Internet access.

While DSL makes use of the telco's existing local telephone infrastructure, **cable Internet access** makes use of the cable television company's existing cable television infrastructure.

A residence obtains cable Internet access from the same company that provides its cable television.

As illustrated in Figure 1.6, fiber optics connect the cable head end to neighborhood-level junctions, from which traditional coaxial cable is then used to reach individual houses and apartments.

Each neighborhood junction typically supports 500 to 5,000 homes.

Because both fiber and coaxial cable are employed in this system, it is often referred to as hybrid fiber coax (HFC).

Cable internet access requires special modems, called cable modems.

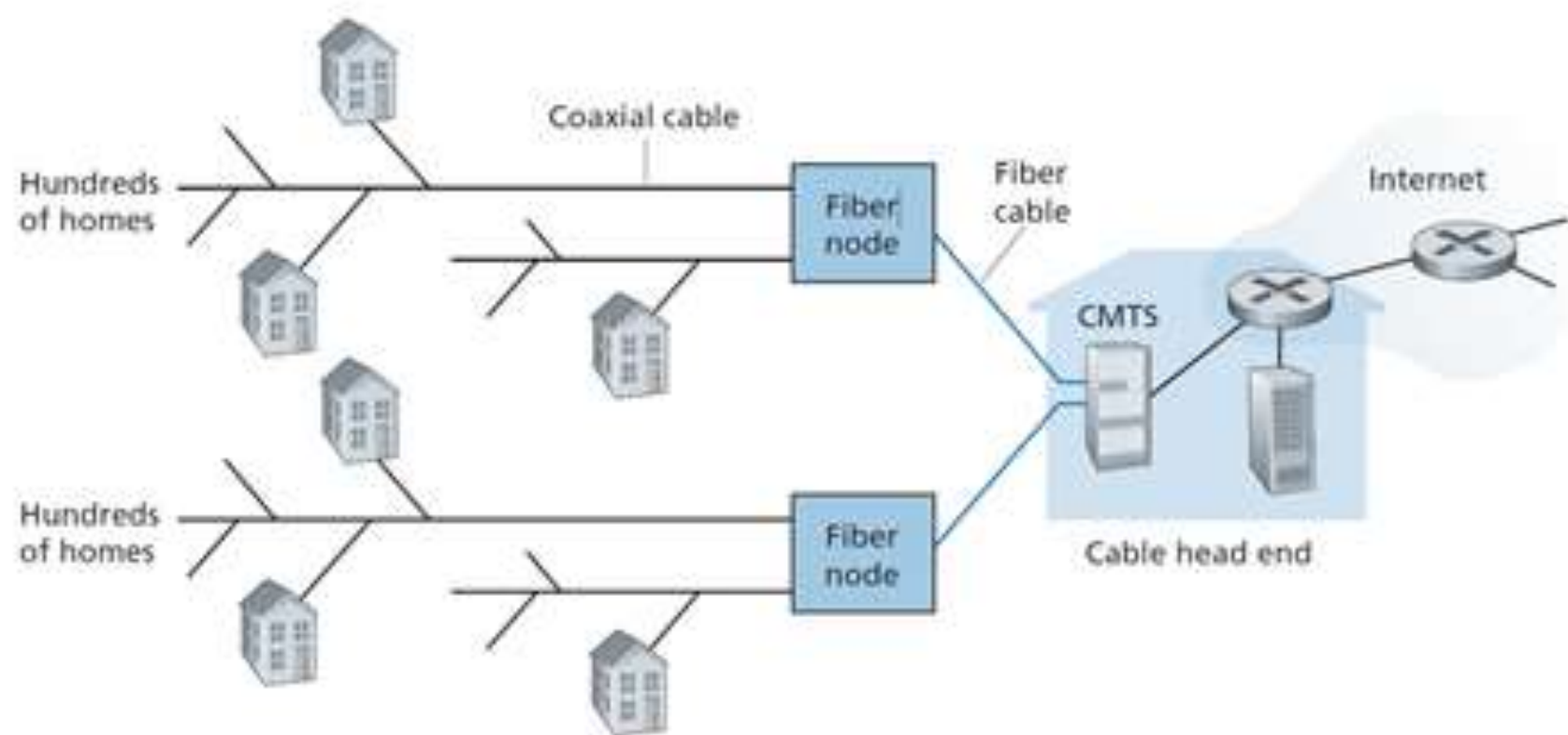


Figure 1.6 • A hybrid fiber-coaxial access network

As with a DSL modem, the cable modem is typically an external device and connects to the home PC through an Ethernet port.

At the cable head end, the cable modem termination system (CMTS) serves a similar function as the DSL network's DSLAM—turning the analog signal sent from the cable modems in many downstream homes back into digital format.

Cable modems divide the HFC network into two channels, a downstream and an upstream channel.

As with DSL, access is typically asymmetric, with the downstream channel typically allocated a higher transmission rate than the upstream channel.

One important characteristic of cable Internet access is that it is a shared broadcast medium.

In particular, every packet sent by the head end travels downstream on every link to every home and every packet sent by a home travels on the upstream channel to the head end.

For this reason, if several users are simultaneously downloading a video file on the downstream channel, the actual rate at which each user receives its video file will be significantly lower than the aggregate cable downstream rate.

On the other hand, if there are only a few active users and they are all Web surfing, then each of the users may actually receive Web pages at the full cable downstream rate, because the users will rarely request a Web page at exactly the same time.

Although DSL and cable networks currently represent the majority of residential broadband access in the United States, an up-and-coming technology that provides even higher speeds is **fiber to the home (FTTH)** [Fiber Broadband 2020]. As the name suggests, the FTTH concept is simple—provide an optical fiber path from the CO directly to the home.

FTTH can potentially provide Internet access rates in the gigabits per second range.

There are several competing technologies for optical distribution from the CO to the homes.

The simplest optical distribution network is called direct fiber,

In addition to DSL, Cable, and FTTH, **5G fixed wireless** is beginning to be deployed.

Access in the Enterprise (and the Home): Ethernet and WiFi

On corporate and university campuses, and increasingly in home settings, a local area network (LAN) is used to connect an end system to the edge router.

Although there are many types of LAN technologies, Ethernet is by far the most prevalent access technology in corporate, university, and home networks.

Ethernet users use twisted-pair copper wire to connect to an Ethernet switch,

The Ethernet switch, or a network of such interconnected switches, is then in turn connected into the larger Internet.

Increasingly, however, people are accessing the Internet wirelessly from laptops, smartphones, tablets, and other “things”.

In a wireless LAN setting, wireless users transmit/receive packets to/from an access point that is connected into the enterprise’s network

A wireless LAN user must typically be within a few tens of meters of the access point.

Wireless LAN access based on IEEE 802.11 technology, more colloquially known as WiFi, is now just about everywhere—universities, business offices, cafes, airports, homes, and even in airplanes.

Even though Ethernet and WiFi access networks were initially deployed in enterprise (corporate, university) settings, they are also common components of home networks.

Many homes combine broadband residential access (that is, cable modems or DSL) with these inexpensive wireless LAN technologies to create powerful home networks

Wide-Area Wireless Access: 3G and LTE 4G and 5G

Mobile devices such as iPhones and Android devices are being used to message, share photos in social networks, make mobile payments, watch movies, stream music, and much more while on the run.

These devices employ the same wireless infrastructure used for cellular telephony to send/receive packets through a base station that is operated by the cellular network provider.

Unlike WiFi, a user need only be within a few tens of kilometers (as opposed to a few tens of meters) of the base station.

Telecommunications companies have made enormous investments in so-called fourth-generation (4G) wireless, which provides real-world download speeds of up to 60 Mbps.

But even higher-speed wide-area access technologies—a fifth-generation (5G) of wide-area wireless networks—are already being deployed.

1.2.2 Physical Media

For example, we said that HFC uses a combination of fiber cable and coaxial cable.

We said that DSL and Ethernet use copper wire.

And we said that mobile access networks use the radio spectrum.

In this subsection, we provide a brief overview of these and other transmission media that are commonly used in the Internet.

In order to define what is meant by a physical medium, let us reflect on the brief life of a bit.

Consider a bit traveling from one end system, through a series of links and routers, to another end system.

This poor bit gets kicked around and transmitted many, many times! The source end system first transmits the bit, and shortly thereafter the first router in the series receives the bit; the first router then transmits the bit, and shortly thereafter the second router receives the bit; and so on.

Thus our bit, when traveling from source to destination, passes through a series of transmitter-receiver pairs.

Thus our bit, when traveling from source to destination, passes through a series of transmitter-receiver pairs.

For each transmitter-receiver pair, the bit is sent by propagating electromagnetic waves or optical pulses across a **physical medium**.

The physical medium can take many shapes and forms and does not have to be of the same type for each transmitter-receiver pair along the path.

Examples of physical media include twisted-pair copper wire, coaxial cable, multimode fiber-optic cable, terrestrial radio spectrum, and satellite radio spectrum.

Physical media fall into two categories: **guided media** and **unguided media**.

With guided media, the waves are guided along a solid medium, such as a fiber-optic cable, a twisted-pair copper wire, or a coaxial cable.

With unguided media, the waves propagate in the atmosphere and in outer space, such as in a wireless LAN or a digital satellite channel.

But before we get into the characteristics of the various media types, let us say a few words about their costs.

The actual cost of the physical link (copper wire, fiberoptic cable, and so on) is often relatively minor compared with other networking costs.

For this reason, many builders install twisted pair, optical fiber, and coaxial cable in every room in a building.

Twisted-Pair Copper Wire

The least expensive and most commonly used guided transmission medium is twisted-pair copper wire.

For over a hundred years it has been used by telephone networks.

In fact, more than 99 percent of the wired connections from the telephone handset to the local telephone switch use twisted-pair copper wire.

Twisted pair consists of two insulated copper wires, each about 1 mm thick, arranged in a regular spiral pattern.

Unshielded twisted pair (UTP) is commonly used for computer networks within a building, that is, for LANs.

Data rates for LANs using twisted pair today range from 10 Mbps to 10 Gbps.

The data rates that can be achieved depend on the thickness of the wire and the distance between transmitter and receiver.

When fiber-optic technology emerged in the 1980s, many people disparaged twisted pair because of its relatively low bit rates.

Some people even felt that fiber-optic technology would completely replace twisted pair.

But twisted pair did not give up so easily.

Modern twisted-pair technology, such as category 6a cable, can achieve data rates of 10 Gbps for distances up to a hundred meters.

In the end, twisted pair has emerged as the dominant solution for high-speed LAN networking.

As discussed earlier, twisted pair is also commonly used for residential Internet access.

We also saw that DSL (digital subscriber line) technology has enabled residential users to access the Internet at tens of Mbps over twisted pair (when users live close to the ISP's central office).

Coaxial Cable

Like twisted pair, coaxial cable consists of two copper conductors, but the two conductors are concentric rather than parallel.

With this construction and special insulation and shielding, coaxial cable can achieve high data transmission rates.

Coaxial cable is quite common in cable television systems.

As we saw earlier, cable television systems have recently been coupled with cable modems to provide residential users with Internet access at rates of hundreds of Mbps.

In cable television and cable Internet access, the transmitter shifts the digital signal to a specific frequency band, and the resulting analog signal is sent from the transmitter to one or more receivers.

Coaxial cable can be used as a guided **shared medium**.

Specifically, a number of end systems can be connected directly to the cable, with each of the end systems receiving whatever is sent by the other end systems.

Fiber Optics

An optical fiber is a thin, flexible medium that conducts pulses of light, with each pulse representing a bit.

A single optical fiber can support tremendous bit rates, up to tens or even hundreds of gigabits per second.

They are immune to electromagnetic interference, have very low signal attenuation up to 100 kilometers, and are very hard to tap.

These characteristics have made fiber optics the preferred long-haul guided transmission media, particularly for overseas links.

Many of the long-distance telephone networks in the United States and elsewhere now use fiber optics exclusively.

Fiber optics is also prevalent in the backbone of the Internet.

However, the high cost of optical devices—such as transmitters, receivers, and switches—has hindered their deployment for short-haul transport, such as in a LAN or into the home in a residential access network.

Terrestrial Radio Channels

Radio channels carry signals in the electromagnetic spectrum.

They are an attractive medium because they require no physical wire to be installed, can penetrate walls, provide connectivity to a mobile user, and can potentially carry a signal for long distances.

The characteristics of a radio channel depend significantly on the propagation environment and the distance over which a signal is to be carried.

Terrestrial radio channels can be broadly classified into three groups:

- those that operate over very short distance (e.g., with one or two meters);

- those that operate in local areas, typically spanning from ten to a few hundred meters;

- and those that operate in the wide area, spanning tens of kilometers.

Personal devices such as wireless headsets, keyboards, and medical devices operate over short distances;

the wireless LAN technologies use local-area radio channels;

the cellular access technologies use wide-area radio channels.

Satellite Radio Channels

A communication satellite links two or more Earth-based microwave transmitter/receivers, known as ground stations.

The satellite receives transmissions on one frequency band, regenerates the signal using a repeater and transmits the signal on another frequency.

Two types of satellites are used in communications: **geostationary satellites** and **low-earth orbiting (LEO) satellites**.

Geostationary satellites permanently remain above the same spot on Earth.

This stationary presence is achieved by placing the satellite in orbit at 36,000 kilometers above Earth's surface.

This huge distance from ground station through satellite back to ground station introduces a substantial signal propagation delay of 280 milliseconds.

Nevertheless, satellite links, which can operate at speeds of hundreds of Mbps, are often used in areas without access to DSL or cable-based Internet access.

LEO satellites are placed much closer to Earth and do not remain permanently above one spot on Earth. They rotate around Earth (just as the Moon does) and may communicate with each other, as well as with ground stations.

1.3 The Network Core

let us now delve more deeply inside the network core—the mesh of packet switches and links that interconnects the Internet's end systems.

1.3.1 Packet Switching

In a network application, end systems exchange **messages** with each other.

Messages can contain anything the application designer wants.

To send a message from a source end system to a destination end system, the source breaks long messages into smaller chunks of data known as **packets**.

Between source and destination, each packet travels through communication links and **packet switches** (for which there are two predominant types, **routers** and **link-layer switches**).

link with transmission rate R bits/sec

Store-and-Forward Transmission

Most packet switches use **store-and-forward transmission** at the inputs to the links.

Store-and-forward transmission means that the packet switch must receive the entire packet before it can begin to transmit the first bit of the packet onto the outbound link.

To explore store-and-forward transmission in more detail, consider a simple network consisting of two end systems connected by a single router, as shown in Figure 1.11.

in this simple example, the router has the rather simple task of transferring a packet from one (input) link to the only other attached link.

In this example, the source has three packets, each consisting of L bits, to send to the destination.

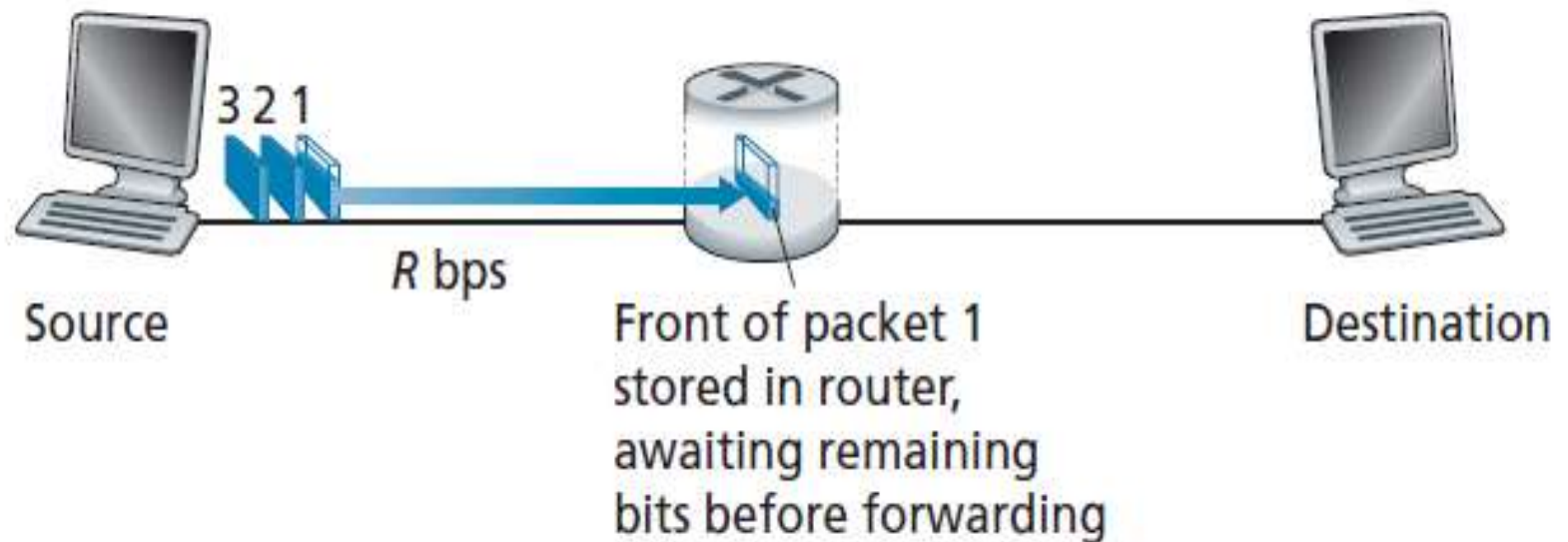


Figure 1.11 ♦ Store-and-forward packet switching

At the snapshot of time shown in Figure 1.11, the source has transmitted some of packet 1, and the front of packet 1 has already arrived at the router.

Because the router employs store-and-forwarding, at this instant of time, the router cannot transmit the bits it has received; instead it must first buffer (i.e., “store”) the packet’s bits.

Only after the router has received *all* of the packet’s bits can it begin to transmit (i.e., “forward”) the packet onto the outbound link.

To gain some insight into store-and-forward transmission, let's now calculate the amount of time that elapses from when the source begins to send the packet until the destination has received the entire packet.

The source begins to transmit at time 0;

at time L/R seconds, the source has transmitted the entire packet, and the entire packet has been received and stored at the router

At time L/R seconds, since the router has just received the entire packet, it can begin to transmit the packet onto the outbound link towards the destination;

at time $2L/R$, the router has transmitted the entire packet, and the entire packet has been received by the destination.

Thus, the total delay is $2L/R$.

Now let's calculate the amount of time that elapses from when the source begins to send the first packet until the destination has received all three packets.

As before, at time L/R , the router begins to forward the first packet.

But also at time L/R the source will begin to send the second packet, since it has just finished sending the entire first packet.

Thus, at time $2L/R$, the destination has received the first packet and the router has received the second packet.

Similarly, at time $3L/R$, the destination has received the first two packets and the router has received the third packet.

Finally, at time $4L/R$ the destination has received all three packets!

Let's now consider the general case of sending one packet from source to destination over a path consisting of N links each of rate R (thus, there are $N-1$ routers between source and destination). Applying the same logic as above, we see that the end-to-end delay is:

$$d_{\text{end-to-end}} = N \frac{L}{R} \quad (1.1)$$

Queuing Delays and Packet Loss

Each packet switch has multiple links attached to it.

For each attached link, the packet switch has an **output buffer** (also called an **output queue**), which stores packets that the router is about to send into that link.

The output buffers play a key role in packet switching.

If an arriving packet needs to be transmitted onto a link but finds the link busy with the transmission of another packet, the arriving packet must wait in the output buffer.

Thus, in addition to the store-and-forward delays, packets suffer output buffer **queuing delays**.

These delays are variable and depend on the level of congestion in the network.

Since the amount of buffer space is finite, an arriving packet may find that the buffer is completely full with other packets waiting for transmission.

In this case, **packet loss** will occur—either the arriving packet or one of the already-queued packets will be dropped.

Figure 1.12 illustrates a simple packet-switched network.

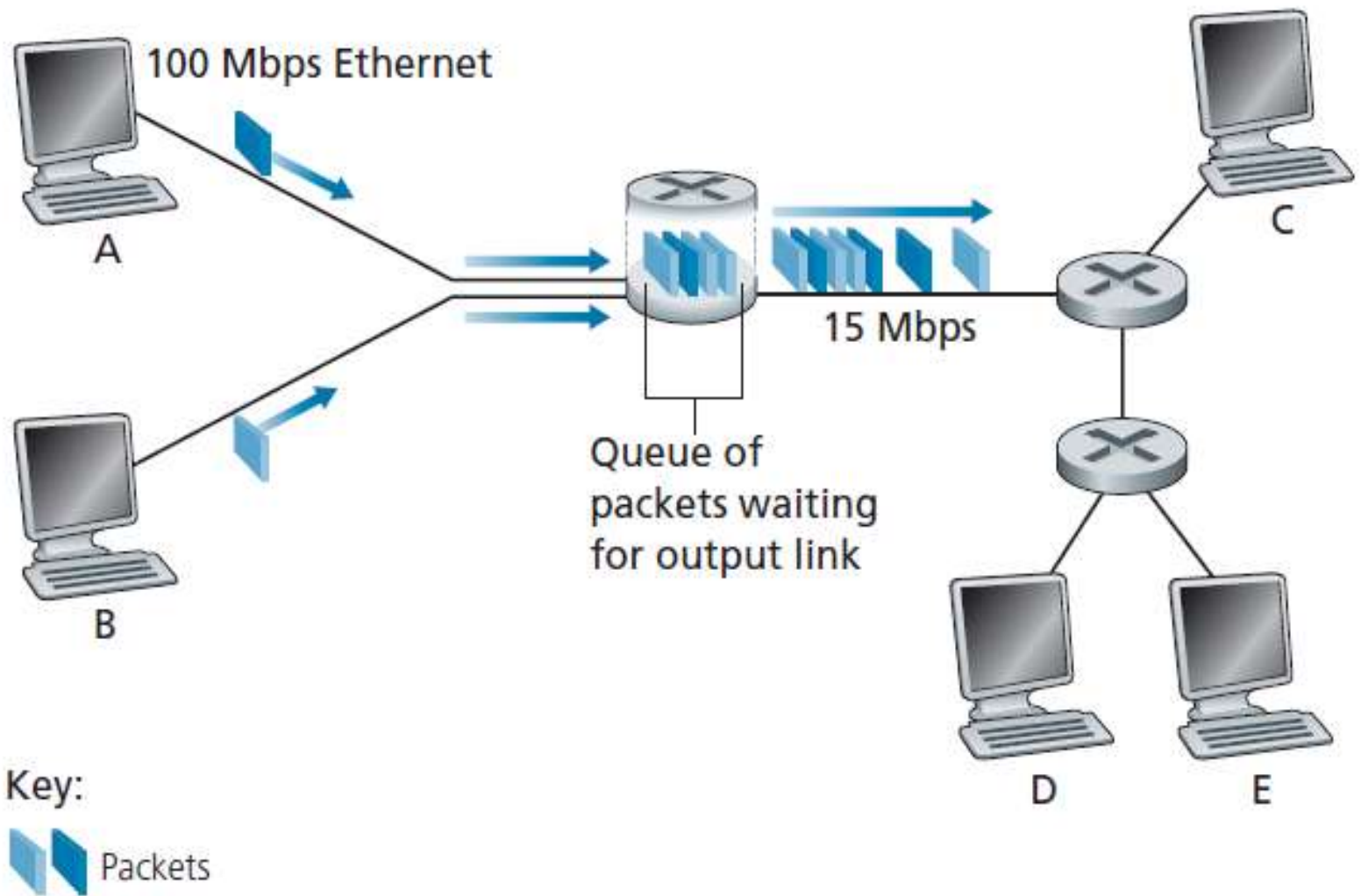


Figure 1.12 ♦ Packet switching

In this figure, all packets have the same length.

Suppose Hosts A and B are sending packets to Host E.

Hosts A and B first send their packets along 100 Mbps Ethernet links to the first router.

The router then directs these packets to the 15 Mbps link.

If, during a short interval of time, the arrival rate of packets to the router exceeds 15 Mbps, congestion will occur at the router as packets queue in the link's output buffer before being transmitted onto the link.

Forwarding Tables and Routing Protocols

Earlier, we said that a router takes a packet arriving on one of its attached communication links and forwards that packet onto another one of its attached communication links.

But how does the router determine which link it should forward the packet onto? Packet forwarding is actually done in different ways in different types of computer networks.

Here, we briefly describe how it is done in the Internet.

In the Internet, every end system has an address called an IP address.

When a source end system wants to send a packet to a destination end system, the source includes the destination's IP address in the packet's header.

When a packet arrives at a router in the network, the router examines a portion of the packet's destination address and forwards the packet to an adjacent router. More specifically, each router has a **forwarding table** that maps destination addresses to that router's outbound links.

When a packet arrives at a router, the router examines the address and searches its forwarding table, using this destination address, to find the appropriate outbound link.

The router then directs the packet to this outbound link.

The end-to-end routing process is analogous to a car driver who does not use maps but instead prefers to ask for directions.

We just learned that a router uses a packet's destination address to index a forwarding table and determine the appropriate outbound link.

we'll note now that the Internet has a number of special **routing protocols** that are used to automatically set the forwarding tables.